GAMIFIED BULIG

BUKIDNON'S UNIFIED LITERACY AND INTERVENTION GATEWAY

TECHNICAL MANUAL

BULIG System Technical Manual

Technologies, system architecture, database schema and data structures of the Bukidnon's Unified Literacy and Intervention Gateway web app.

Version 208 | November 2026 | For developers and ICT coordinators



DepEd Division of Bukidnon · BULIG · Building Up Literacy, Inspiring Growth

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1 · SYSTEM OVERVIEW

BULIG at a glance

BULIG turns the DepEd BULIG literacy modules into a web app that pupils use on a phone, tablet or computer, at school or at home. Teachers follow every pupil, and the Division administrator manages schools, teachers and data.

9
reading levels (1, 2A, 2B, 3A, 3B, 4, 5, 6, 7)

58
database tables · 327 columns

3
roles: pupil, teacher, administrator

58
PHP feature files in `app/`

29
browser scripts in `public/assets/`

576
automatic tests

Who uses BULIG

ROLE	WHAT THEY DO
Pupil	Takes the reading check at the first sign-in, learns level by level on the road map, reads with Max the Reading Coach, earns XP, stars, ranks and badges, and asks for help.
Teacher	Adds pupils (one or a whole class, DepEd SF1 works), sets starting levels and Reading Support, verifies readings, sees class results, runs Class Demo and prints tickets, report cards and certificates.
Administrator	Manages schools and teachers, sees Division reports and app ratings, answers bug reports, sends messages, makes backups and cleans old data.

2 · TECHNOLOGIES USED

The technology stack

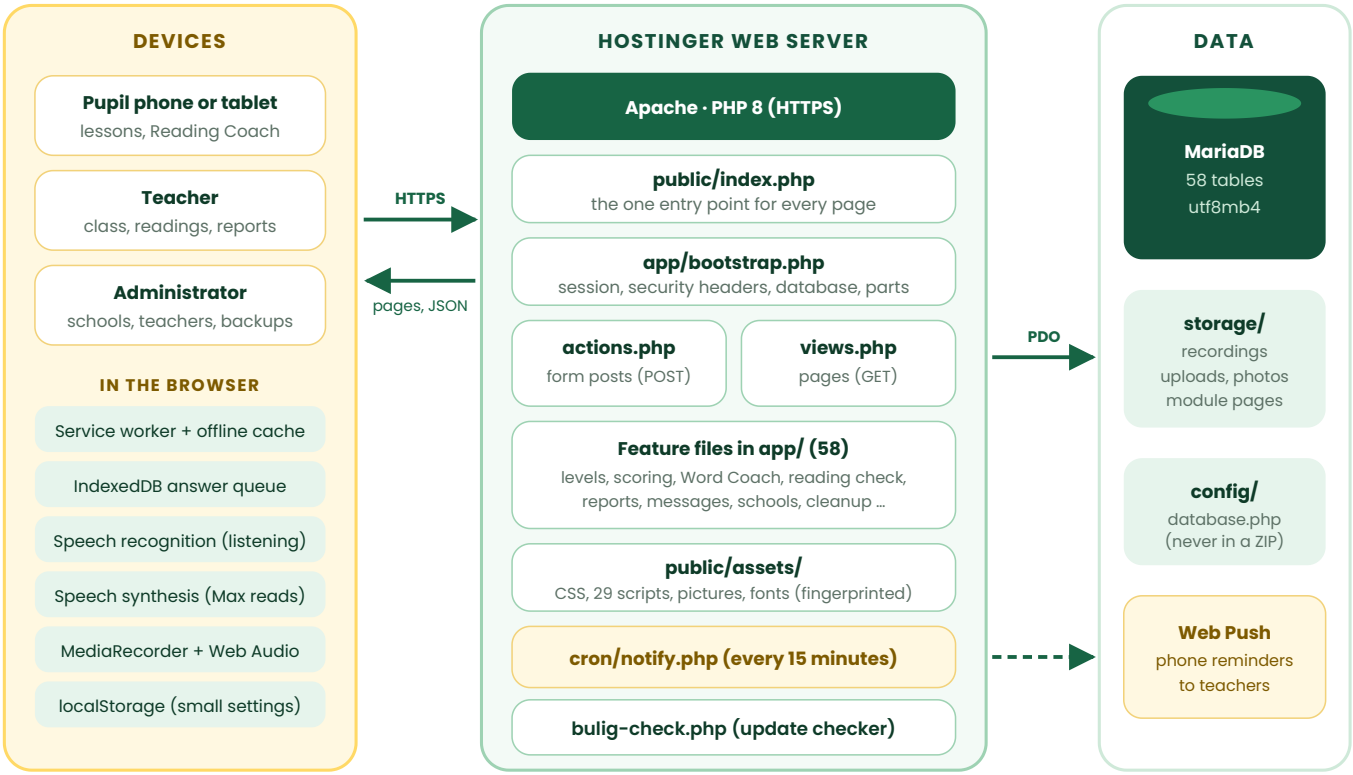
BULIG uses plain, well-known technology with no framework and no build step, so it runs on low-cost shared hosting and on old phones.

LAYER	TECHNOLOGY	USED FOR
Server language	PHP 8 (strict types)	Every page and form. One entry point, `public/index.php`. No framework.
Database	MySQL or MariaDB 10.x (InnoDB, utf8mb4)	All accounts, lessons, answers and results. Accessed with PDO prepared statements.
Web server	Apache on Hostinger (HTTPS)	Serves `public/`. The project sits in `public_html`.
Pages	Hand-written HTML5 and CSS3	One design system in `theme.css`: green and ivory colours, Poppins font, light and dark mode.
Browser code	Plain JavaScript (ES2020)	29 small scripts, loaded with `defer`, one per feature.
Listening	Web Speech API: SpeechRecognition	Hears the pupil read; each word turns green when said right.
Reading aloud	Web Speech API: SpeechSynthesis	Max, Jarvis and the friends read questions and stories.
Recording	MediaRecorder, getUserMedia, Web Audio	Read-aloud recordings for the teacher; the live microphone meter; sound effects made in code.
Offline	Service Worker, Cache Storage, IndexedDB	Install as an app, open saved levels offline, keep answers until the internet comes back.
Reminders	Web Push (VAPID) + cron	`cron/notify.php` runs every 15 minutes and sends phone reminders.
Libraries	jsQR 1.4, qrcode-generator 2.0, html2canvas 1.4	Read and print QR sign-in tickets; save boards as pictures.
Pictures and fonts	WebP, PNG, SVG icons, Poppins (WOFF2, local)	Small, fast pictures; every icon is an SVG (no emoji).
Tools and tests	Python 3, Playwright (Chromium), Node.js	Content builders, the update builder, 576 automatic tests and page checks.

Why no framework? Shared hosting, slow school internet and old phones. Plain PHP and JavaScript keep pages small, updates simple (copy files from a ZIP) and the code easy for a new developer to read.

How the parts fit together

A classic three-tier web app: the browser on the pupil's or teacher's device, the PHP application on the web server, and the MariaDB database with files in storage.



Browser (client)

Draws the pages, listens and records with the device microphone, reads aloud, keeps offline copies and queues answers in IndexedDB until they can be sent.

Application (server)

PHP checks the session and role, runs the action or builds the page, scores answers with the module answer keys, and writes to the database.

Data

MariaDB holds all records. Recordings, uploaded photos and module page pictures are files in `storage/`, served only through PHP after an access check.

Scheduled task

`cron/notify.php` sends queued phone reminders (Web Push) to teachers and pupils who allowed them.

The life of a request

1. The browser asks for `index.php?page=…` (a page) or sends a form with an `action` (POST).
2. `app/bootstrap.php` starts the session (HttpOnly, Secure, SameSite cookies), sends the security headers (Content-Security-Policy), connects to the database and loads every feature file listed in `BULIG_PARTS`.
3. For a POST, `action()` in `app/actions.php` checks the CSRF token, the role (`require_role`) and ownership (`own_pupil`), changes the data inside a transaction, writes the activity log and redirects (Post, Redirect, Get).
4. For a GET, the page function draws HTML with `head()`, the page body and `foot()`. Every value is escaped with `e()`.
5. Before the page is sent, BULIG's own titles get Title Case. Text from the database (names, sections, subjects) stays as written.
6. The browser loads the fingerprinted CSS and scripts (for example `coach-1a2b3c4d5e.js`), so an update never shows an old cached file.

Lesson pages by level

LEVEL	DRAWN BY	HOW PUPILS ANSWER
Level 1 · Oral Language	`level1.php`, `l1.js`	Listen, speak and record, act out, draw.
2A, 2B · Phonological Awareness	`level2a.php`, `l2a.js`	Card by card: letter tiles, word chips, pictures, say it.
3A, 3B · Word Recognition	`level2a.php` (card engine)	Word cards turn green as each word is read.
Level 4 · Fluency	`level4.php`, `coach.js`	Read the passage with the Reading Coach; Phil-IRI miscue scoring.
5, 6, 7 · Comprehension and Love for Reading	card engine + `scoring.php`	Choose, type, match, order, write, read aloud.

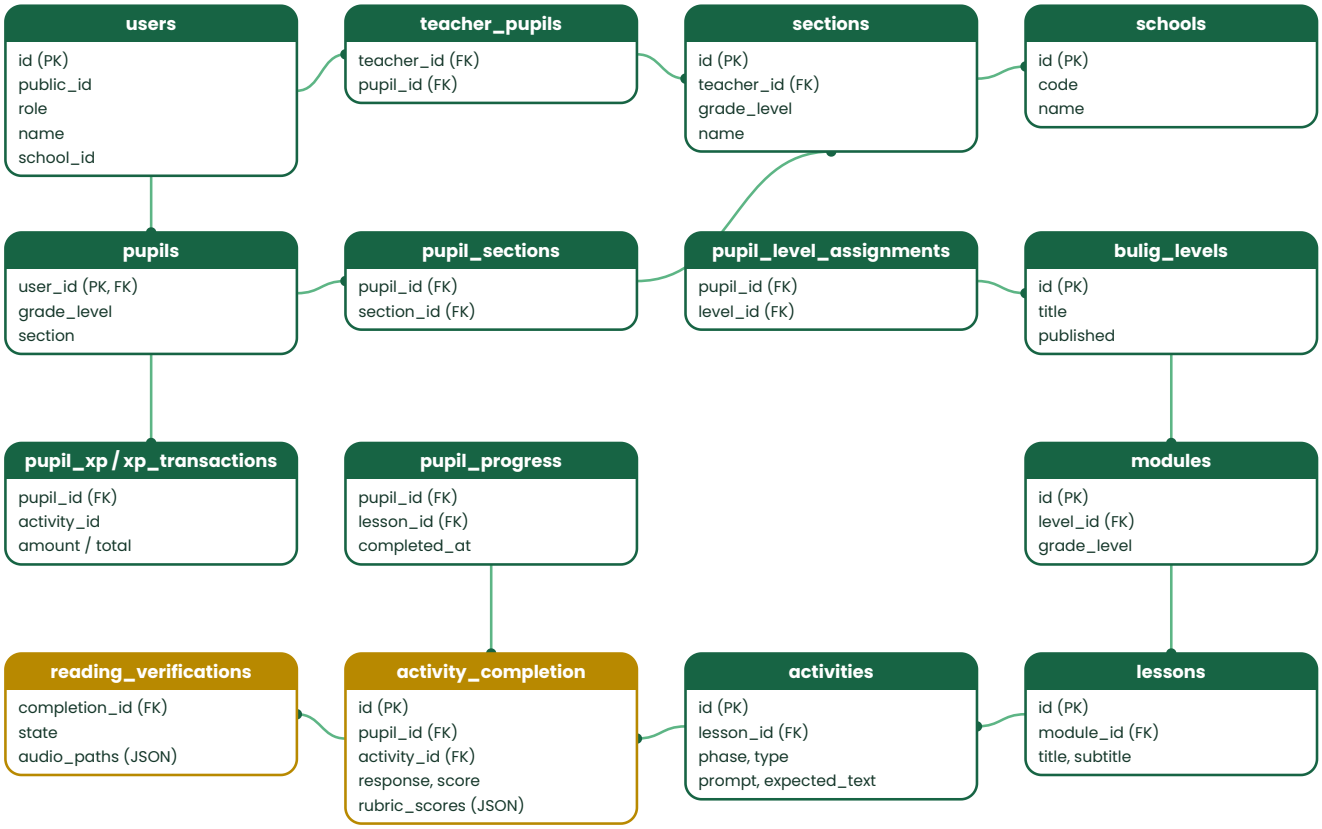
5 · FOLDERS AND FILES

Where things are

FOLDER	WHAT IS IN IT
app/	All PHP code. `bootstrap.php` (start), `actions.php` (forms), `views.php` (pages), and one file per feature: `scoring.php`, `coach.php`, `level4.php`, `reading_check.php`, `reading_support.php`, `reading_verification.php`, `class_done.php`, `messages.php`, `schools.php`, `jeepy.php` (road map) and more.
public/	The only folder the browser can reach: `index.php`, `sw.js` (offline), `bulig-check.php` (update checker) and `assets/` (CSS, scripts, pictures, fonts).
database/	`fresh_install_v2.sql` for a new install, `migrations/` (38 numbered update files, safe to run again) and the lesson content files (JSON).
storage/	Recordings, uploaded photos and the pages of the original modules. Not reachable directly from the web.
config/	`database.php`: database name, user and password. Never put the live one in an update ZIP.
cron/	`notify.php`, the scheduled task for phone reminders.
tools/	Content builders, `update_checker.py` (file fingerprints), `make_update.py`, `shrink_images.py`, `create_admin.php`.
tests/	`run_tests.py` (576 tests) and browser checks.
live-update/	Update ZIPs and SQL files for the live site.
docs/	Developer guide and content notes for each level.

The main relationships

The database has 58 tables and 327 columns. The diagram shows the core: accounts, classes, lesson content and answers. Green boxes are core tables; yellow boxes hold the pupils' answers and readings.



One account table

Pupils, teachers and administrators are all rows in `users` (column `role`). Extra details live in `pupils`, `teachers`, `user_names` and `pupil_details`.

Content is data

Levels, modules, lessons and activities are rows, built from the DepEd modules by the content tools, so content can be fixed without changing code.

One answer per activity

`activity_completion` has one row per pupil and activity. Earlier copies go to `response_history`.

Safe to update

Migrations use `CREATE TABLE IF NOT EXISTS` and checked `ALTER TABLE`, and `schema_migrations` records what ran.

7 · DATABASE SCHEMA

Every table

All 58 tables in six groups, with their columns, types and keys. PK primary key · FK foreign key · required NOT NULL. The file name shows where the table is made.

People and accounts

16 tables

users

fresh_install_v2.sql

Every account: pupils, teachers and administrators. Holds the sign-in ID, role, name and password hash.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
public_id	varchar(30)	required
role	enum('admin','teacher','pupil')	required
name	varchar(150)	required
password_hash	varchar(255)	required
active	boolean	required
avatar_path	varchar(255)	
created_at	timestamp	
school_id	int unsigned	

pupils

fresh_install_v2.sql

Pupil details that belong to learning: grade level and section name.

COLUMN	TYPE	NOTES
user_id	bigint unsigned	PK FK → users
grade_level	tinyint unsigned	FK → grade_levels required
section	varchar(80)	required

teachers

fresh_install_v2.sql

Marks a user as a teacher.

COLUMN	TYPE	NOTES
user_id	bigint unsigned	PK FK → users

admins

fresh_install_v2.sql

Marks a user as an administrator.

COLUMN	TYPE	NOTES
user_id	bigint unsigned	PK FK → users

admin_security

014_admin_pin.sql

The administrator PIN (hashed), failed tries and lock time.

COLUMN	TYPE	NOTES
user_id	bigint unsigned	PK, FK → users
pin_hash	varchar(255)	
failed_pins	tinyint unsigned	required
locked_until	datetime	
updated_at	timestamp	

user_names

030_name_parts.sql

First, middle and last name and extension (Jr., Sr.) for DepEd name format.

COLUMN	TYPE	NOTES
user_id	bigint unsigned	PK, FK → users, required
first_name	varchar(60)	required
middle_name	varchar(60)	
last_name	varchar(60)	required
name_ext	varchar(4)	

pupil_details

006_pupil_details.sql

Sex and LRN (learner reference number) of each pupil.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	PK, FK → pupils
sex	enum('male','female')	
lrn	varchar(12) character set ascii collate ascii_bin	

account_id_history

027_account_ids.sql

Old and new sign-in IDs when an ID format changes, so old IDs still work.

COLUMN	TYPE	NOTES
id	int unsigned	PK, required
user_id	bigint unsigned	FK → users, required
old_id	varchar(30)	required
new_id	varchar(30)	required
changed_at	timestamp	

id_sequences

fresh_install_v2.sql

The next number for each kind of sign-in ID.

COLUMN	TYPE	NOTES
kind	varchar(20)	PK
next_value	bigint unsigned	required

saved_logins

016_saved_logins.sql

“Remember me” tokens (hashed) for quick sign-in on a shared device.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
user_id	bigint unsigned	FK → users, required
token_hash	char(64)	required
created_at	timestamp	
last_used_at	datetime	
expires_at	datetime	required

login_attempts

fresh_install_v2.sql

Wrong password tries (hashed ID), used to slow down guessing.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
identifier_hash	char(64)	required
attempted_at	timestamp	

schools

031_schools.sql

Schools in the Division, with code, name, district, head and logo.

COLUMN	TYPE	NOTES
id	int unsigned	PK
code	varchar(10)	required
name	varchar(150)	required
school_no	varchar(40)	required
district	varchar(150)	required
head	varchar(150)	required
logo	varchar(255)	required
active	tinyint(1)	required
created_at	timestamp	

sections

fresh_install_v2.sql

Class sections made by teachers (grade and name).

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
teacher_id	bigint unsigned	FK → teachers, required
grade_level	tinyint unsigned	FK → grade_levels, required
name	varchar(80)	required
created_at	timestamp	
archived_at	datetime	

pupil_sections

fresh_install_v2.sql

Which section each pupil is in.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	PK, FK → pupils
section_id	bigint unsigned	FK → sections, required

teacher_pupils

fresh_install_v2.sql

Which pupils belong to which teacher.

COLUMN	TYPE	NOTES
teacher_id	bigint unsigned	FK → teachers, required
pupil_id	bigint unsigned	PK, FK → pupils

grade_levels

fresh_install_v2.sql

Grades 1 to 6.

COLUMN	TYPE	NOTES
id	tinyint unsigned	PK
name	varchar(30)	required

Lesson content

10 tables

bulig_levels

fresh_install_v2.sql

The 9 BULIG levels (1, 2A, 2B, 3A, 3B, 4, 5, 6, 7) and whether each is published.

COLUMN	TYPE	NOTES
id	int unsigned	PK
title	varchar(150)	required
published	boolean	required

modules

fresh_install_v2.sql

One module per level (Levels 4 to 7: one per grade).

COLUMN	TYPE	NOTES
id	int unsigned	PK
level_id	int unsigned	FK → bulig_levels, required
title	varchar(200)	required
grade_level	tinyint unsigned	FK → grade_levels

lessons

fresh_install_v2.sql

Lessons in each module, with objectives, teacher guide and source pages.

COLUMN	TYPE	NOTES
id	int unsigned	PK
module_id	int unsigned	FK → modules, required
position	int	required
title	varchar(255)	required
subtitle	varchar(255)	
objectives	text	required
teacher_guide	longtext	required
source_pages	json	required
image_path	varchar(255)	
published	boolean	required
created_at	timestamp	

activities

fresh_install_v2.sql

Every activity screen: phase (pre, learn, post), type, instructions, prompt, pictures, expected text and XP.

COLUMN	TYPE	NOTES
id	int unsigned	PK
lesson_id	int unsigned	FK → lessons, required
assessment_id	int unsigned	FK → assessments
phase	enum('pre','learn','post')	required
position	int	required
title	varchar(255)	required
type	enum('open','sentence','reading','physical','group','drawing','choice','exact','reference')	required
instructions	text	required
prompt	text	required
image_path	varchar(255)	
image_paths	json	
narration	text	required
expected_text	text	
xp_reward	int unsigned	required
source_page	int	required
source_excerpt	text	required
published	boolean	required
revision	int unsigned	required
response_mode	enum('answer','none','perform','drawing')	

assessments

fresh_install_v2.sql

Pre-tests and post-tests that group activities.

COLUMN	TYPE	NOTES
id	int unsigned	PK
lesson_id	int unsigned	FK → lessons, required
kind	enum('pre','post')	required
title	varchar(255)	required
rubric	text	required
source_page	int	required
source_note	text	

assessment_questions

fresh_install_v2.sql

Questions that belong to an assessment.

COLUMN	TYPE	NOTES
assessment_id	int unsigned	FK → assessments, required
question_id	bigint unsigned	FK → questions, required

questions

fresh_install_v2.sql

Questions for an activity (older content format).

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
activity_id	int unsigned	FK → activities, required
content	text	required
grading	enum('teacher','exact')	required

answers

fresh_install_v2.sql

Possible answers for a question (older content format).

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
question_id	bigint unsigned	FK → questions, required
content	text	required
correct	boolean	required

source_pages

fresh_install_v2.sql

Text and pictures of each page of the original DepEd modules.

COLUMN	TYPE	NOTES
page_number	int unsigned	PK
text_content	longtext	required
image_path	varchar(255)	required
images	json	required

content_issues

fresh_install_v2.sql

Notes about mistakes found in the source modules and how BULIG handled them.

COLUMN	TYPE	NOTES
id	int unsigned	PK
description	text	required
resolution	text	
acknowledged_by	bigint unsigned	FK → admins
acknowledged_at	datetime	

Learning, answers and reading

11 tables

activity_completion

fresh_install_v2.sql

A pupil's answer to one activity: response, drawing, transcript, audio, status, score and teacher feedback.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → pupils, required
activity_id	int unsigned	FK → activities, required
response	text	required
drawing	longtext	
transcript	text	
prompt_snapshot	text	required
activity_revision	int unsigned	required
status	enum('submitted','approved','retry')	required
score	decimal(5,2)	
max_score	decimal(5,2)	
feedback	text	
reviewed_by	bigint unsigned	FK → users
submitted_at	datetime	required
reviewed_at	datetime	
audio_path	varchar(160)	
rubric_scores	text	
audio_paths	text	

activity_drafts

fresh_install_v2.sql

Unsent answers saved while the pupil works.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	FK → pupils, required
activity_id	int unsigned	FK → activities, required
response	text	required
drawing	longtext	
updated_at	timestamp	

response_history

fresh_install_v2.sql

Earlier copies of an answer and every teacher decision on it.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
completion_id	bigint unsigned	FK → activity_completion, required
response	text	required
status	varchar(20)	required
actor_id	bigint unsigned	FK → users, required
created_at	timestamp	

assessment_attempts

fresh_install_v2.sql

A pupil's pre-test or post-test attempt and its score.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → pupils, required
assessment_id	int unsigned	FK → assessments, required
status	enum('in_progress','submitted','reviewed')	required
score	decimal(8,2)	
max_score	decimal(8,2)	
completed_at	datetime	

pupil_progress

fresh_install_v2.sql

Lessons a pupil started and finished.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	FK → pupils, required
lesson_id	int unsigned	FK → lessons, required
last_activity_id	int unsigned	FK → activities
completed_at	datetime	
updated_at	timestamp	

pupil_level_assignments

fresh_install_v2.sql

The starting level the teacher (or the reading check) chose.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → pupils, required
level_id	int unsigned	FK → bulig_levels, required
assigned_by	bigint unsigned	FK → teachers, required
assigned_at	timestamp	

learning_days

fresh_install_v2.sql

Each day a pupil learned (for streaks and the calendar).

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	FK → pupils, required
day	date	required

reading_checks

028_reading_check.sql

The first-sign-in reading check: parts done, result and the level it placed.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → users, required
grade	tinyint unsigned	required
status	enum('running','done')	required
state	longtext	
bulig_level	int unsigned	
placed_level	int unsigned	
changed_by	bigint unsigned	
teacher_note	text	
talk_rating	varchar(20)	
official_tool	varchar(10)	
official_result	varchar(30)	
started_at	datetime	required
finished_at	datetime	
updated_at	datetime	required

reading_assessments

fresh_install_v2.sql

Older reading word-match results.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → pupils, required
activity_id	int unsigned	FK → activities, required
transcript	text	required
expected_text	text	required
word_match_percent	decimal(5,2)	required
detail	json	required
created_at	timestamp	

reading_verifications

035_reading_verification.sql

Read-aloud recordings waiting for the teacher to verify, with the decision and any XP reversed.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK required
completion_id	bigint unsigned	FK → activity_completion required
pupil_id	bigint unsigned	FK → users required
activity_id	int unsigned	required
state	enum('pending','missing','verified', 'rejected','superseded')	required
audio_paths	mediumtext	
prompt_snapshot	mediumtext	
response_snapshot	mediumtext	
score	decimal(10,2)	
max_score	decimal(10,2)	
reason	varchar(3000)	
reviewed_by	bigint unsigned	FK → users
reviewed_at	datetime	
xp_reversed	int unsigned	required
created_at	datetime	required

help_requests

025_help_requests.sql

"I need help" taps from pupils, with the reason.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → users required
lesson_id	bigint unsigned	
activity_id	bigint unsigned	
reason	varchar(20)	required
created_at	datetime	required

Rewards and motivation

8 tables

pupil_xp

fresh_install_v2.sql

Total XP of each pupil.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	PK FK → pupils
total	int unsigned	required

xp_transactions

fresh_install_v2.sql

XP earned for each activity (one row per activity).

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
pupil_id	bigint unsigned	FK → pupils, required
activity_id	int unsigned	FK → activities, required
amount	int unsigned	required
created_at	timestamp	

badges

fresh_install_v2.sql

Badges that can be earned.

COLUMN	TYPE	NOTES
id	int unsigned	PK
title	varchar(100)	required
description	varchar(255)	required
rule_type	enum('activity','lesson','xp','streak','level','reading','perfect')	required
threshold_value	int unsigned	required
active	boolean	required

pupil_badges

fresh_install_v2.sql

Badges each pupil earned.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	FK → pupils, required
badge_id	int unsigned	FK → badges, required
earned_at	timestamp	

rewards

fresh_install_v2.sql

Rewards unlocked at an XP total.

COLUMN	TYPE	NOTES
id	int unsigned	PK
title	varchar(100)	required
required_xp	int unsigned	required
active	boolean	required

pupil_rewards

fresh_install_v2.sql

Rewards each pupil unlocked.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	FK → pupils, required
reward_id	int unsigned	FK → rewards, required
earned_at	timestamp	

pupil_streaks

fresh_install_v2.sql

Current and longest learning streak.

COLUMN	TYPE	NOTES
pupil_id	bigint unsigned	PK, FK → pupils
current_streak	int unsigned	required
longest_streak	int unsigned	required
last_activity_date	date	

pupil_cards

018_pupil_cards.sql

Encrypted starter password for printing sign-in tickets.

COLUMN	TYPE	NOTES
user_id	bigint unsigned	PK, FK → users, required
starter_enc	varchar(255)	
issued_at	timestamp	

Messages, reminders and feedback

10 tables

notifications

024_notifications.sql

Bell notifications for every user.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
user_id	bigint unsigned	FK → users, required
kind	varchar(30)	required
title	varchar(200)	required
body	varchar(300)	required
url	varchar(250)	required
created_at	datetime	required
read_at	datetime	

admin_messages

033_admin_messages.sql

Messages sent by the administrator.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
sender_id	bigint unsigned	FK → users
send_token	char(48)	required
subject	varchar(150)	required
body	text	required
created_at	datetime	required

message_recipients

033_admin_messages.sql

Who received each message and when they read it.

COLUMN	TYPE	NOTES
message_id	bigint unsigned	FK → admin_messages required
user_id	bigint unsigned	FK → users required
read_at	datetime	

message_replies

037_message_replies.sql

Replies to administrator messages.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
message_id	bigint unsigned	required
user_id	bigint unsigned	required
author_id	bigint unsigned	FK → users
body	text	required
created_at	datetime	required
read_at	datetime	

push_queue

017_push_reminders.sql

Phone reminders waiting to be sent by the scheduled task.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
user_id	bigint unsigned	FK → users, required
kind	varchar(20)	required
title	varchar(120)	required
body	varchar(200)	required
url	varchar(200)	required
created_at	timestamp	
sent_at	datetime	

push_subscriptions

017_push_reminders.sql

Devices that accept phone reminders (Web Push).

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
user_id	bigint unsigned	FK → users, required
endpoint_hash	char(64)	required
endpoint	varchar(500)	required
p256dh	varchar(120)	required
auth	varchar(60)	required
prefs	varchar(255)	
last_sent_on	date	
created_at	timestamp	

bug_reports

034_bug_workflow.sql

Problems reported through “I need help › Something is not working”.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
reporter_id	bigint unsigned	FK → users, required
status	enum('open','processing','fixed','rejected')	required
summary	varchar(300)	required
context	text	required
penalty_applied	boolean	required
deducted_xp	int unsigned	required
created_at	datetime	required
updated_at	datetime	required

bug_report_updates

034_bug_workflow.sql

Status changes and replies on a bug report.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
report_id	bigint unsigned	FK → bug_reports, required
admin_id	bigint unsigned	FK → users
status	enum('open','processing','fixed','rejected')	required
body	text	required
deducted_xp	int unsigned	required
notification_id	bigint unsigned	FK → notifications
popup_seen_at	datetime	
created_at	datetime	required

bug_report_links

034_bug_workflow.sql

Links between new reports and older notifications.

COLUMN	TYPE	NOTES
notification_id	bigint unsigned	PK, FK → notifications
report_id	bigint unsigned	FK → bug_reports, required

app_ratings

029_app_ratings.sql

Stars from pupils and the 5-question evaluation from teachers.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
user_id	bigint unsigned	FK → users, required
role	enum('pupil','teacher')	required
stars	tinyint unsigned	required
overall	decimal(3,1)	required
answers	varchar(40)	
comment	text	
created_at	datetime	required

System

3 tables

settings

fresh_install_v2.sql

Key and value pairs: school info, feature switches and small per-user flags (see Data structures).

COLUMN	TYPE	NOTES
setting_key	varchar(80)	PK
setting_value	text	required

audit_log

fresh_install_v2.sql

Every important action: who did what and when.

COLUMN	TYPE	NOTES
id	bigint unsigned	PK
actor_id	bigint unsigned	FK → users
action	varchar(100)	required
details	text	required
created_at	timestamp	

schema_migrations

fresh_install_v2.sql

Which update files have been run on this database.

COLUMN	TYPE	NOTES
version	varchar(80)	PK
applied_at	timestamp	

Formats inside the data

Some columns hold structured text or JSON. These are the formats a developer needs to know.

Card activity answer (`activity_completion.response`)

```
BULIG card answers
1: bed
2: (said) the cat is small
3: picture 2
Coach: 1=83|sleep yard; 2=100|
Tries: 1=2; 3=1
```

One line per card. The **Coach** line holds each card's Word Coach result: `percent|words to practise` (`na` when the device could not listen). **Tries** counts attempts.

Scores and reading results (`rubric_scores`, JSON)

```
{"wc":{"p":92,"w":["sleep","yard"]},
 "l4":{"m":
[1,0,1,0,0,0,0],"total":2,"words":27,"score":92.6,"level":"Instructional","secs":41,"wpm":36},
 "teacher_score":3}
```

`wc` Word Coach result · `l4` Level 4 Phil-IRI miscues (mispronunciation, omission, substitution, insertion, repetition, transposition, reversal) · `teacher_score` a score the teacher set.

Recordings (`audio_paths`, JSON)

```
{"0":"recordings/3/568-1730612345.webm","2":"recordings/3/412-1730612399.m4a"}
```

Card number to file in `storage/`. Card 0 is the whole activity.

Settings keys (`settings` table)

KEY	MEANING
speech_<pupil id>	Reading Support: `accent` (Accent-friendly) or `support` (Speech support). No row means Standard.
whatsnew_<version>_<user id>	The user has seen that update popup.
tour_seen_, jarvis_met_, friends_met_…	One-time welcomes and tours already shown.
last_backup_at	When the last backup was downloaded (Clear Old Data needs one today).
school info keys	School name, head, school year and logos for reports and report cards.

On the device (browser storage)

NAME	HOLDS
IndexedDB bulig-offline	Answers waiting to upload, with the time they were given.
Cache bulig-off-<pupil>-L<level>	A level saved for offline use.
localStorage wcprog:…	Where an interrupted reading stopped (for “Welcome back”, kept 24 hours).
localStorage bulig-sfx, bulig-voice, bulig-mic-ok	Sound effects on or off, the chosen voice, microphone help already shown.

How BULIG hears and scores reading

The Reading Coach (Word Coach)

1. **Max reads** the text slowly while the words light up (the pupil may skip).
2. **You read**: speech recognition hears the pupil; each word turns green with a chime. The live meter shows the microphone hears them.
3. **Fix the missed words**: each missed word gets 5 hearts. The score is 1 point for a word read right the first time, less after hearts are lost.
4. **Summary**: stars and the words to practise. The reading and recording go to the teacher. On iPhone and iPad, the recording is a second short step.

Matching words

BULIG compares sounds, not only letters: `then` and `den`, `10` and `ten`, `wet` and `wed` count as the same. **Accent-friendly** pupils also get i/e, o/u, p/f, b/v, s/z/sh, th as t, and a last sound not said, except in lessons that teach a vowel. **Speech support** pupils (for example cleft palate) are never marked wrong: the teacher listens and scores.

Level 4 fluency (Phil-IRI)

`l4_miscues()` aligns what was heard with the passage (dynamic programming) and counts the seven miscue kinds. Score = (words - miscues) ÷ words; the level is Independent (97% and up), Instructional (90%) or Frustration.

Answer keys and the post-test gate

`scoring.php` checks answers with the module keys (`answer-keys.json`, `answer-keys-2.json`). A pupil opens the next level after 80% on the post-test (`PT_PASS`).

Teacher review

Read-aloud recordings go to `reading_verifications`. Verify Readings lists them, with readings BULIG could not score at the top (“Waiting for You”). Rejecting asks for a reread and takes back that activity’s XP once.

Keeping pupils and data safe

Passwords and PIN

Hashed with `password_hash` (bcrypt). Pupils get an easy starter password on a printed ticket. The administrator also needs a PIN, locked after wrong tries.

Guessing limits

`login_attempts` slows down repeated wrong passwords. Sign-in IDs are hashed in that table.

Sessions

HttpOnly and Secure cookies with SameSite. Saved sign-ins use hashed, expiring tokens.

Forms

Every form has a CSRF token. Every action checks the role and that the pupil belongs to the teacher.

Database

Only PDO prepared statements; changes that touch many tables run in a transaction.

Content-Security-Policy

`default-src 'self'`: no outside scripts or inline code; styles and scripts come only from BULIG.

Recordings and photos

Kept outside `public/` and served only to the pupil's own teacher (and the pupil).

Activity log

`audit_log` records sign-ins, changes, deletions, merges, cleanups and undos.

Privacy: Reading Support and the red plus or blue bubble beside a name are seen only by the pupil's teacher and the administrator, never by classmates or on class screens.

Learning without internet

1. BULIG can be added to the home screen (a Progressive Web App with `sw.js`).
2. **Save for offline** stores a whole level (pages and pictures) in the device cache.
3. Answers given offline wait in IndexedDB and upload by themselves when the internet is back. The teacher sees which pupils still have answers waiting.
4. Teachers can save Class Demo levels to teach without internet.

Speed on old phones

Fingerprinted assets are cached for a long time; pictures are WebP and shrunk; scripts are small and deferred; background drawings are plain CSS gradients with no animation.

Sending an update to the live site

1. Make the change and run `python3 tools/update_checker.py`. It updates the file fingerprints in `public/bulig-check.php`.
2. Run the tests (section 13).
3. Build a ZIP in `live-update/` with only the changed files, plus `public/bulig-check.php`. Never include `config/database.php`.
4. Upload the ZIP to Hostinger's File Manager in `public_html` and extract it, replacing files.
5. If the update has an SQL file, import it in phpMyAdmin. Migrations are safe to run again.
6. Open Admin › Checkup: it compares every file with its fingerprint and lists anything missing or old.

Order matters: upload ZIPs in the order they were made. A later ZIP has the newest copy of a file.

New install

Create a database, import `database/fresh_install_v2.sql` and the content files, fill in `config/database.php`, create the first administrator with `tools/create_admin.php`, and add the cron job for `cron/notify.php` every 15 minutes.

Checking that everything works

CHECK	HOW	WHAT IT COVERS
Automatic tests	`python3 tests/run_tests.py`	576 checks: install and updates, sign-in, roles, every level, scoring, reading check, Level 4 miscues, messages, bug reports, offline sync, backups and security.
Browser checks	`node tests/audit-regressions.cjs`	Uploads, downloads, offline queue and loading.
Layout checks	Playwright at 360, 390, 768 and 1280 px, light and dark	Overlaps, cut-off text and sideways scrolling on every page and activity.
Diagnostics	Admin › Checkup (`bulig-check.php`)	Files, tables and settings on the live site.

Keeping the data healthy

Backups

Print & Download has an Excel backup (pupils, teachers, sections, results) and a full database file (.sql). Make one at least once a month and before every update.

Data cleanup

Join pupils who have two accounts, archive last year's sections, and clear old records (activity log, read notifications, sign-in attempts, help requests, earlier answer copies, messages). Pupils' answers, scores and XP are never removed.

New school year

Settings › New school year moves pupils up a grade after a backup.

Content issues

Mistakes found in the source modules are listed on the Content issues page with how BULIG handled them.

Glossary

Word Coach / Reading Coach	Max helps the pupil read; words turn green when said right.
Reading check	The short placement test at the first sign-in.
Phil-IRI	The Philippine Informal Reading Inventory, used for Level 4 scoring.
CRLA	Comprehensive Rapid Literacy Assessment; an official result a teacher can enter.
SF1	DepEd School Form 1, the class list BULIG can import.
Fingerprint	A short code from a file's content, used in its name so phones never use an old copy.
Migration	A numbered SQL update file in `database/migrations`.